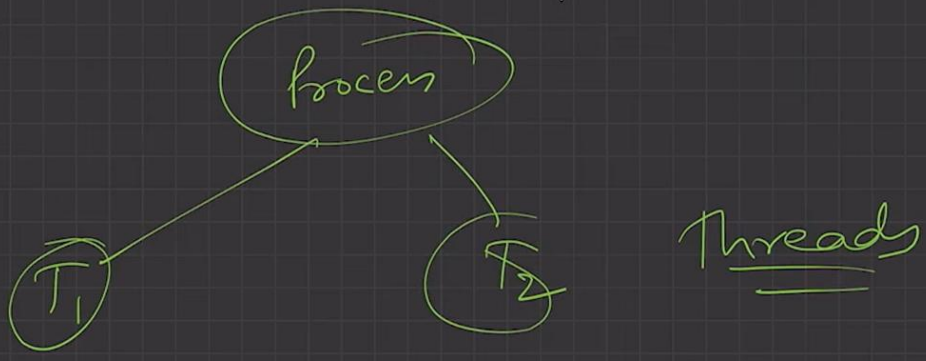
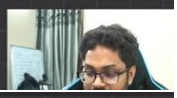
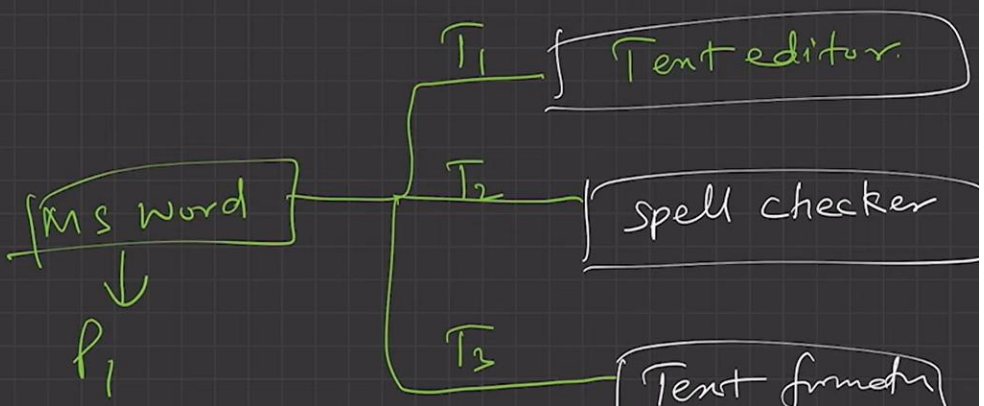
Lec-15

Concurrency :- Multiple instruction at the same time.

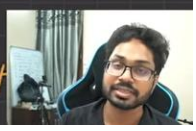


Threads





Text editor → spell checker → Text



Process

Single process
P_1

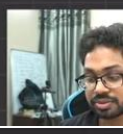
Address space

Threads

P_1

T_1
T_2
T_3

addrn space.



Process

Stack
↓
heap
data
text

Thread

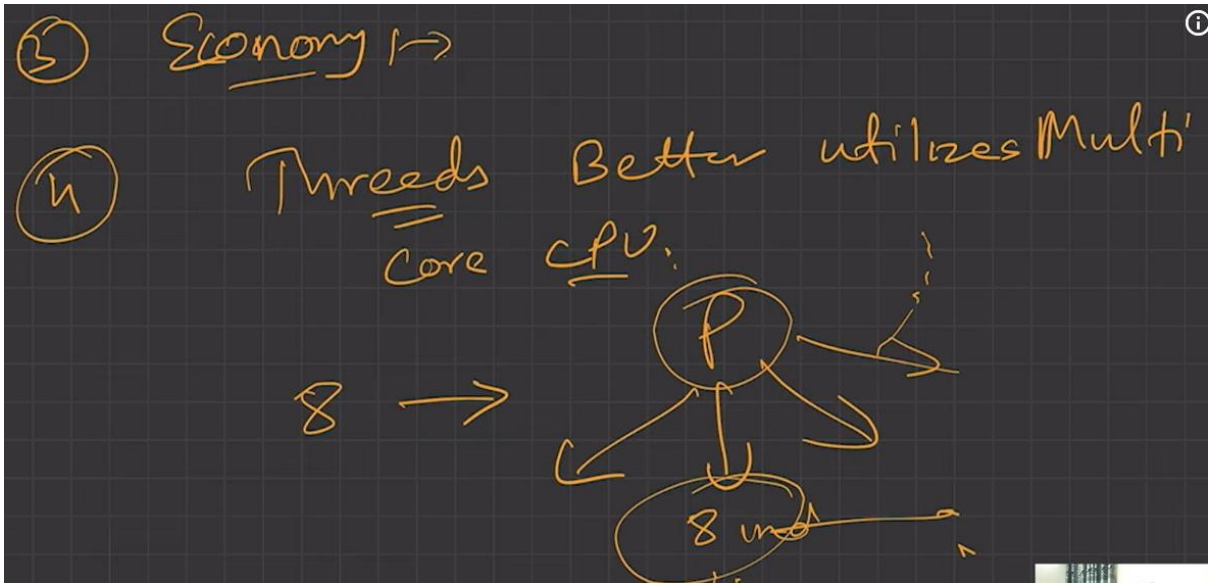
T_1 ← Stack 1
T_2 ← Stack 2
heap
data
text

Benefits of MT

(1) Responsiveness
=
Interactive

(2) Resource sharing →

address space.
Shared.



thread_cpp.cpp

Users > lakshaykumar > Desktop > thread_cpp.cpp > main()

```cpp
#include <iostream>
#include <thread>
#include <unistd.h>

using namespace std;

void taskA()
{
    for (int i = 0; i < 10; ++i)
    {
        sleep(1);
        printf("TaskA: %d\n", i);
        fflush(stdout);
    }
}

void taskB()
{
    for (int i = 0; i < 10; ++i)
    {
        sleep(1);
        printf("TaskB: %d\n", i);
        fflush(stdout);
    }
}

int main()
{
    thread t1(taskA);
    thread t2(taskB);

    t1.join();
    t2.join();
    return 0;
}
```

TERMINAL — zsh - Desktop

```
lakshaykumar@192 Desktop % g++ thread_cpp.cpp
lakshaykumar@192 Desktop % ./a.out
libc++abi: terminating
zsh: abort      ./a.out
lakshaykumar@192 Desktop % g++ thread_cpp.cpp
lakshaykumar@192 Desktop % ./a.out
TaskA: 0
TaskB: 0
TaskA: 1
TaskB: 1
TaskA: 2
TaskB: 2
TaskA: 3
TaskB: 3
TaskA: 4
TaskB: 4
TaskA: 5
TaskB: 5
TaskB: 6
TaskA: 6
TaskA: 7
TaskB: 7
TaskA: 8
TaskB: 8
TaskA: 9
TaskB: 9
lakshaykumar@192 Desktop %
```

